WEB-BASED QUEUING MANAGEMENT SYSTEM WITH PREDICTIVE WAITING MECHANISM AND ADAPTIVE COUNTER ALLOCATION

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RFT
MMB
CNAE
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JAVR

DEDICATION

This study is dedicated to our families for their endless love, encouragement, and sacrifices that have given us the strength to complete this journey.

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ABSTRACT

ROJEAN F. TANDAS, MANRIJAN M. BARBOSA, CARREN NICOLE A. ESTRELLES, JOHN DAVID R. GARCIA, JADE ALLYSA V. RULLODA (2027). **Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation.** Bachelor of Science in Computer Science. Don Mariano Marcos Memorial State University South La Union Campus, College of Computer Science, Agoo, La Union.

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Manual queuing in local government offices often causes prolonged waiting times, physical congestion, and inefficient counter utilization. This study proposes the design and implementation of a Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union. The system integrates QR-code ticket issuance, Phantom Queue Simulation for dynamic wait-time estimation, dynamic priority scoring, automated SMS alerts, and adaptive counter rebalancing between Peace and Crunch modes. Employing developmental and descriptive research designs under an Agile methodology, the system undergoes iterative sprint cycles encompassing unit, integration, and live alpha/beta testing to evaluate cloud synchronization and wait-time accuracy. System quality will be assessed by BPL-LIEO personnel, transacting citizens, and IT experts using an evaluation instrument grounded in the ISO/IEC 25010 software quality standards.

Keywords: *Web-Based Queuing Management, Predictive Waiting Mechanism, Adaptive Counter Allocation, Phantom Queue Simulation, ISO/IEC 25010, BPL-LIEO Rosario La Union*

Chapter 1

INTRODUCTION

Situation Analysis

Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEOs) play an essential role in municipal governance by facilitating new business registrations, permit renewals, and related transactions. During peak renewal periods, these offices accommodate massive influxes of clients. When relying on manual or purely localized queuing methods, this volume often results in crowded waiting areas, client frustration due to uncertain wait times, and immense pressure on frontline personnel to manually balance different transaction types. To deliver efficient public service, an intelligent and communicative queue management system is required.

The architecture of a queue significantly impacts service efficiency. Ainbinder, Temnikov, and Allalouf (2024) emphasized that queue design and routing policies directly dictate waiting times and workload distribution. Furthermore, Lee et al. (2024) demonstrated that flexible priority schemes allow service resources to be dynamically allocated based on current queue conditions, improving efficiency while maintaining fairness. These principles highlight the necessity for a queue management system that does not merely issue sequential numbers, but rather employs smart algorithms to auto-balance workloads and prioritize specific client groups (such as PWDs or returning citizens) in real time.

In the Asian context, the transition towards digital and real-time queue management has proven effective in mitigating crowding. Chantima and Pochan (2023) demonstrated

the importance of updating queue information based on continuous real-time conditions rather than static sequences. Similarly, Santoso et al. (2024) and Fauzi and Suhaimi (2026) showcased the utility of web-based systems integrating QR-code access and live waiting-status tracking. These technological leaps are highly relevant, as they support the architectural shift from isolated local networks (LAN) to web-based platforms, allowing clients to track their queue status via their mobile devices rather than remaining confined to a physical waiting room.

In the Philippines, the push for frontline digitalization is largely driven by the Anti-Red Tape Authority (ARTA), which has mandated the implementation of the electronic Business One-Stop Shop (eBOSS). While the eBOSS platform streamlines the filing and payment of permits (Dela Merced, Balmeo, & De Castro, 2024), it primarily addresses the transaction process itself, not the physical flow and waiting experience of the clients inside the municipal hall. Studies such as those by Espinosa, Serrano, and Balmes (2024) and Alegado and Himo (2022) confirm that even with computerized records, there remains a critical need for dedicated queue management systems to organize physical client flow, reduce employee workload, and broadcast transparent queue data.

This need is evident at the regional and local levels in Region I. A relevant study was conducted by Rodriguez et al. (2025) in Alaminos City, Pangasinan. The study aimed to streamline government service processes, minimize waiting time, and improve service efficiency and customer experience. Using a mixed-methods approach involving waiting-time analysis, system-performance measures, customer satisfaction surveys, and staff interviews, the researchers reported improvements in waiting time, customer satisfaction, workflow efficiency, and the organization and transparency of the Queuing process. The

system also allowed personnel to manage transactions and prioritize urgent cases. In addition, the municipality of Bacnotan in La Union was reported as the second LGU in the Ilocos Region, after the City of San Fernando, La Union, to pass ARTA's assessment of fully streamlined and digitalized business permit application-to-issuance processes which included a unified application form, electronic tax bills or orders of payment, online uploading of requirements, electronic permits, integrated online payment, and courier integration.

While neighboring municipalities like Alaminos City (Rodriguez et al., 2025) and Bacnotan have made strides in digital Queuing and eBOSS compliance, local government units retain distinct internal workflows. For instance, the municipalities of Agoo and Sudipen maintain separate procedures and document assessments for new applications versus renewals. This distinction dictates that a queue system cannot be "one-size-fits-all"; it must be heavily adapted to the specific transaction types, assessment stages, and operational requirements of the target LGU.

At the BPL-LIEO of Rosario, La Union, the current workflow presents distinct challenges that standard Queuing cannot resolve. Clients often step away from the waiting area, leading to skipped numbers, "no-shows," and the need to process "return-tomorrow" transactions. Without a way to proactively communicate with clients, staff face bottlenecks when clients are absent when called.

Therefore, the gap addressed by this study is the urgent need for a communicative, adaptive queuing solution. By developing a Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation, this study seeks to eliminate the physical confines of the waiting room. Through the integration of QR code

mobile tracking, automated SMS and web push notifications, and auto-balancing algorithms, the system aims to provide clients with transparency regarding their wait times while giving BPL-LIEO personnel an automated tool to manage service flow efficiently.

Statement of Objectives

This study aims to design and implement a Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union to streamline client flow, eliminate waiting room congestion, and optimize counter service efficiency.

Specifically, the study seeks to achieve the following objectives:

1. To design and develop a Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation that includes:

    1.1 **Ticket Generator Module:** Generates and prints physical tickets with unique sequence numbers and scannable QR codes based on selected transaction types (*New Business, Renewal, Retirement*) and priority classifications (*Regular, Senior Citizen, PWD, Pregnant, Returning*).

    1.2 **Smart Queue & Adaptive Counter Allocation Engine:** Ranks waiting clients using the dynamic Smart Score ($S(t)$) formula and dynamically reallocates active service windows between Peace Mode and Crunch Mode based on queue pressure and SLA thresholds.

1.3 **Predictive Wait-Time Mechanism:** Estimates dynamic client waiting times using the Phantom Queue Simulation model ($E[W]$) with abandonment discounting based on active counter capacity and rolling average transaction times.

1.4 **Staff-Controlled Dashboard & Public Monitor Display:** Enables counter personnel to call, recall, defer, postpone, or complete transactions, synchronized in real time with a public Smart TV display featuring voice synthesis and audio chimes.

1.5 **Mobile Tracking & Multi-Channel Notification Module:** Allows clients to monitor live queue rank and estimated wait times on their mobile devices via QR scan, accompanied by automated SMS text alerts (near-turn, called, recalled).

1.6 **Statistical Reporting & SLA Analytics Module:** Tracks daily/historical transaction volume, counter utilization, drop-off rates, SLA compliance, and staff performance metrics.

2. To test and improve the system's performance, real-time cloud data synchronization, and overall functionality through iterative alpha and beta testing; and
3. To assess the level of quality of the developed web-based system using the ISO/IEC 25010 Software Quality Standards in terms of functional suitability, performance efficiency, usability, reliability, security, maintainability, and portability.

Time and Place of Study

This study will be conducted at the Don Mariano Marcos Memorial State University – South La Union Campus, College of Computer Science, located in Consolacion, Agoo, La Union, and the Municipality of Rosario, specifically at the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO), Rosario, La Union for the School Year 2026–2027.

Definition of Terms

For a better understanding of this research report, the following terms are operationally defined:

Adaptive Counter Allocation refers to the system's dynamic mechanism that reallocates active service windows between Peace Mode and Crunch Mode based on queue pressure, transaction workload, and SLA thresholds.

Auto-Balancing refers to the system's intelligent algorithm that automatically manages queue workloads across available service windows based on Service Level Agreement (SLA) thresholds, dynamic wait times, and priority weights.

Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) refers to the local government office of LGU Rosario, La Union responsible for business registrations, permit renewals, investment promotions, and licensing-related transactions. It serves as the primary beneficiary and implementation site of the proposed queue management system.

Client refers to an individual or business representative who visits the BPL-LIEO to inquire, submit requirements, process a new business permit, renew an existing permit, or perform another supported business-related transaction.

Current Queue Number refers to the queue number presently being served by the assigned service window. It is displayed in real time to inform waiting clients.

Done refers to the queue status assigned to a ticket when the client's transaction has been completed by the assigned personnel.

Mobile Tracking refers to the ability of clients to scan a printed QR code and monitor their real-time queue position and status via their smartphones, eliminating the need to remain in the physical waiting area.

Monitor Display refers to the public-facing screen used to display the currently served queue ticket and succeeding queue information so that waiting clients can determine when they should proceed for service.

Multi-Channel Notifications refers to the automated alerts sent to clients via SMS text messages and web push notifications as their turn to be served approaches or if their ticket status changes.

Predictive Waiting Mechanism refers to the system's mathematical wait-time estimation model utilizing Phantom Queue Simulation, active counter capacity, rolling average transaction durations, and dynamic abandonment discounting.

Priority Classification refers to the specific classification of a client (e.g., PWD, Senior Citizen, Pregnant, Returning) that dictates their weight in the queue, granting them appropriately elevated priority over regular transactions.

Priority/Returning Client refers to individuals who require expedited processing, including Senior Citizens, Persons with Disabilities (PWDs), pregnant women, or individuals returning to the BPL-LIEO to complete previously unfinished transactions. Within the proposed system, these clients are assigned elevated priority weights to ensure they are accommodated efficiently without causing queue starvation for regular clients.

Queue History refers to the collection of recorded queue transactions from the system maintained by authorized personnel for monitoring, statistics, reporting, and future reference.

Queue Management System refers to the automated system developed in this study that manages the issuance of queue numbers, monitors queue status in real time, and supports staff-controlled queue operations to improve service efficiency in the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO).

Queue Number refers to the unique number automatically generated by the system and assigned to a client according to the selected transaction type. It determines the client's order of service.

Queue Status refers to the current condition of a queue ticket within the proposed system. The statuses include Waiting, Serving, Done, Return Tomorrow, and Priority Waiting.

Real-Time Status Updates refer to the live display of queue information, including the current queue number, the next queue number, and changes in queue status as transactions are processed.

Service Window refers to the designated counter where authorized personnel process clients' business permit and licensing transactions after their queue numbers are called.

Serving refers to the queue status assigned to a ticket when the client is currently being accommodated or processed by the assigned personnel.

Staff-Controlled Queue Operations refer to the functions available to authorized personnel for managing the queue. These include calling the next queue number, skipping, removing, and completing queue transactions to maintain an organized client flow.

Transaction Type refers to the category of service selected by the client, such as new business registration, business permit renewal, or other licensing-related transactions. It determines the appropriate queue assignment and service process.

Waiting refers to the queue status assigned to a newly generated ticket that has not yet been called by authorized personnel.

Chapter 2

METHODOLOGY

Research Design

The researchers will utilize a combination of developmental and descriptive research designs. Developmental research design will be employed following the framework cited in Torion and Bacio (2024). Developmental research is a systematic process of creating, implementing, and assessing educational programs, processes, or products to ensure their internal consistency and effectiveness. This research design will be employed to systematically design, develop, and evaluate the Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO). This approach is appropriate as the primary goal of the study is to create a functional technological solution to an existing problem.

Descriptive research design will be used during the evaluation phase to assess the system's quality. Descriptive research design is defined as a method used to describe the characteristics, the current state, or the features of a population, a situation, or phenomenon as it exists. It aims to gather information about "what is" by using tools like questionnaires or interviews to capture people's views, behaviors, or conditions (Sahin & Mete, 2021). This study will involve surveying the end-users (BPL-LIEO staff and citizens) as well as IT Experts/System Evaluators to describe their experience and gather quantitative feedback on the system's performance.

Materials and Procedures

The following hardware and software tools will be used in the development of the system:

Hardware

Hardware includes laptops/desktops for development and staff usage, a standard thermal/ink printer for ticket and QR code generation, a Smart TV or monitor for the queue display, and internet-connected routers for web synchronization.

Software

For Frontend UI & State Management, software includes React.js powered by Vite (for fast bundling), Tailwind CSS (for styling), Framer Motion (for UI animations), Recharts (for analytics dashboards), Lucide React (for iconography), React Query (for efficient data fetching), and QR Code React (for mobile ticket generation).

For Backend & Security, software includes Node.js with the Express.js framework for handling RESTful APIs. Security and validation are handled via JSON Web Tokens (JWT), bcrypt for password hashing, and Zod for strict schema validation. API documentation is generated using Swagger UI.

For Database and Cloud Sync, software includes Prisma ORM with SQLite (configured with Write-Ahead Logging [WAL] mode for concurrent read/write transactions) for resilient local edge persistence, tightly integrated with Firebase Firestore for real-time cloud data synchronization.

For Real-time Communication and Notifications, software includes Socket.io for live bidirectional event broadcasting between dashboards and displays, the Vonage Server SDK for SMS messaging, and Web Push libraries for browser-based notifications.

For the Development Tools, software includes Visual Studio Code (VS Code), Git for version control, and modern web browsers (Chrome/Edge/Safari) for testing.

Mathematical Queuing Model and Adaptive Algorithms

To address the complexities of prioritizing different client types (e.g., PWDs, Senior Citizens) while maintaining fairness for regular clients, the system deviates from a standard First-In-First-Out (FIFO) queue. Instead, it utilizes a proprietary Smart Queue Engine powered by dynamic mathematical modeling. The system computes a real-time Smart Score $S(t)$ for every ticket t in the queue, continuously re-sorting clients based on priority weights, wait times, and Service Level Agreement (SLA) thresholds.

1. **Adaptive Auto-Balancing Formula**

   The Smart Score for any given ticket is calculated using the following equation:

$$S(t) = P_{base} + \left(W_{mins} \times R_{aging}\right) + P_{SLA} + P_{skip} + P_{zipper}$$

   Where:

- $\boldsymbol{P_{base}}$: The base priority weight assigned to the client's demographic (e.g., elevated weights for Priority Groups such as Senior Citizens or Pregnant Women).

  Regular = 0, Returning Client = 20, Senior Citizen / PWD / Pregnant = 50

- $\boldsymbol{W_{mins}}$: The elapsed waiting time in minutes since the ticket was generated.

- $\boldsymbol{R_{aging}}$: The configurable aging rate. This ensures that "Regular" tickets passively gain points over time, preventing permanent queue starvation by an influx of priority clients.

  1.5 points per minute. *(This ensures a regular client waiting 34 minutes naturally overtakes a newly arrived priority client, balancing fairness with priority).*

- $\boldsymbol{P_{SLA}}$: The SLA Panic Bonus (+100 points). If W_{mins}exceeds the office's predefined acceptable wait time threshold, the ticket is forcefully expedited.

  +100 points, triggered when $W_{\text{mins}} \geq 30$ minutes(the ARTA threshold).

- $\boldsymbol{P_{skip}}$: The Stoplight Skip Limit Bonus (+500 points). Applied when a client has been deferred or skipped a maximum allowable number of times.

+500 points, applied when skipped $\geq$ 2times.

- $\boldsymbol{P_{zipper}}$: The Zipper Force Bonus (+1000 points). A strict starvation-prevention rule triggered when the last Nconsecutive clients served were exclusively priority types. This forces the oldest regular ticket to instantly bypass the queue, ensuring a fair "zipper" merge of priority and regular clients.

  +1000 points, triggered when $N = 3$consecutive priority clients have been served.

2. **Predictive Wait Time with Phantom Simulation**

   To provide accurate waiting time estimates for incoming clients, the system executes a "Phantom Queue Simulation." An invisible phantom ticket is injected into the current queue matrix to calculate its true rank (R_{true}) based on the $S(t)$algorithm.

   Because physical queues experience "no-shows" as wait times increase, the system applies an **Abandonment Discount** (D) to account for human behavior:

$$D = \min\left(0.30, \left\lfloor \frac{R_{true} - 1}{10} \right\rfloor \times 0.05\right)$$

   (This applies a 5% wait-time reduction for every 10 clients ahead in the queue, up to a maximum 30% discount).

Finally, the **Expected Wait Time** $E[W]$in minutes is dynamically calculated across all actively manned service counters (C) and the rolling average service time per transaction (T_{avg}):

$$E[W] = \max\left(1, \text{round}\left(\frac{R_{true} \times (1 - D)}{C} \times T_{avg}\right)\right)$$

To prevent division-by-zero exceptions during off-peak or transition periods when zero staff counters are logged in ($C = 0$), the denominator defaults to an effective minimum of $C_{\text{eff}} = 1$ (or displays 'Service Window Closed') to ensure system stability.

3. Adaptive Counter Allocation Algorithm (Peace vs. Crunch Mode)

To optimize counter utilization during fluctuating citizen arrivals, the system dynamically orchestrates service counters between two operating states:

- Peace Mode: Operating under normal traffic where queue lengths across all services are below SLA thresholds. All active service counters remain in generalized mode, serving tickets across New, Renewal, and Retirement queues based purely on Smart Score $S(t)$.
- Crunch Mode: Automatically triggered when any transaction category exceeds its predefined queue threshold ($Q_threshold$) or maximum acceptable wait time

(SLA_limit) . The engine dynamically calculates proportional counter distribution:

$$C_s = \max\left(C_{\min}, \text{round}\left(C_{\text{total}} \times \frac{L_s}{\sum L_i}\right)\right)$$

Where:

- C_s: The number of active service counters assigned to transaction type s.
- C_total: Total active, manned counters currently logged into the system.
- L_s: The weighted queue load for service s, determined by waiting tickets multiplied by average service duration ($L_s = Q_s * T_avg, s$).
- $sum(L_i)$: The aggregate queue workload across all active transaction categories.
- C_min: The starvation floor constraint ($\mathrm{C_{min}} = 1$) is active when $\mathrm{C_{total}} \geq 3$. If active staff falls below the number of active services ($\mathrm{C_{total}} < 3$), Crunch Mode is suppressed and the engine automatically operates in Peace Mode (shared counters serving all transaction types) to maintain physical feasibility.

A 3-minute hysteresis deadband is applied to prevent rapid, disruptive counter reassignments during brief queue fluctuations.

By establishing these calculatable metrics, the system moves beyond arbitrary Queuing, providing a statistically sound methodology for adaptive counter allocation, wait time prediction, and fair priority inclusion.

Software Development Life Cycle (SDLC)

The study will adopt the Agile Methodology as its software development life cycle model. In Agile development, software is constructed incrementally through iterative cycles (sprints), allowing the development team to adapt to changing requirements, continuously integrate feedback, and systematically test features. The project will execute the following Agile phases:

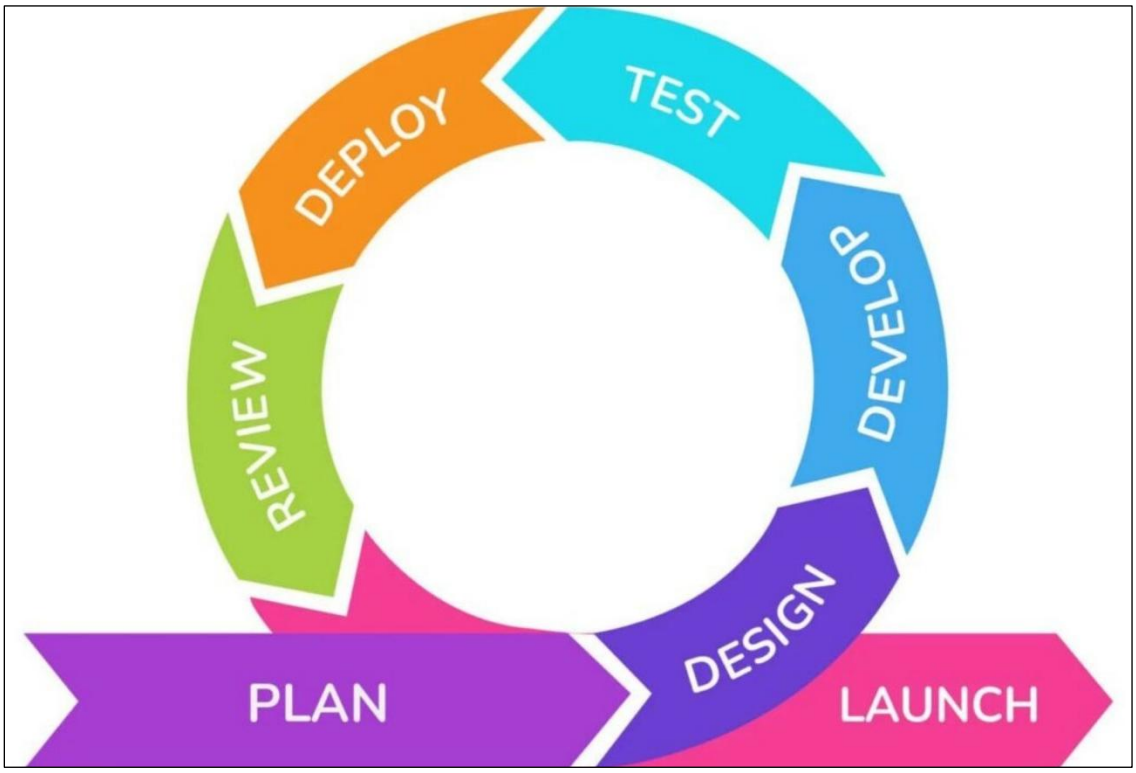


Fig. 1. Agile Software Development Life Cycle Model (cited in Nordin et al., 2025)

Plan. In Agile, this phase involves identifying user needs, defining project scope and objectives, translating requirements into prioritized features and estimating the workload for the development process (Nordin et al., 2025). The researchers conducted stakeholder interviews and observational workflow analysis with BPL-LIEO personnel to identify operational bottlenecks in manual queuing. Key functional requirements were established, including transaction-based ticket issuance with QR codes, live counter calling, priority management for returning clients, smart auto-balancing logic, automated multi-channel notifications (SMS and Web Push), and comprehensive statistical reporting.

Design. This phase is when the designers, architects, and developers work collaboratively to develop the software architecture and the corresponding UI and UX for the product (Milne, 2021). In this study, the researchers will design the system architecture using Entity-Relationship Diagrams (ERD) for relational data persistence, Data Flow Diagrams (DFD Levels 0 and 1) for information routing across the cloud and local network, Use-Case Diagrams for actor-role boundaries, and interactive wireframes for the Staff Dashboard, TV Display, Admin Analytics panels, and the Mobile Client Tracker interface.

Develop. This is when teams work in accordance with the designed software architecture (Milne, 2021). In this study, the researchers utilized a modern JavaScript tech stack comprising React.js powered by Vite and Tailwind CSS for the responsive frontend user interfaces, Node.js with the Express.js framework for backend RESTful APIs, Prisma ORM with SQLite for reliable local data persistence along with Firebase/Firestore for seamless cloud integration. Furthermore, Vonage and Web Push libraries were integrated for notifications, and Socket.io was implemented for bidirectional, real-time event broadcasting across staff dashboards, public displays, and mobile devices. Security was integrated via JWT and bcrypt.

Test. This phase involves evaluating the system's accuracy and compatibility by performing several types of testing (Nordin et al., 2025). Agile emphasizes continuous testing alongside development to ensure code quality, socket event stability, and immediate defect resolution (LTS Group, 2023). The researchers will perform unit testing on individual API endpoints, integration testing on WebSocket emitters and Cloud sync triggers, and stress testing under simulated concurrent client loads.

Furthermore, formal Alpha Testing will be conducted in a simulated lab environment by peer developers, followed by Beta Testing conducted with actual BPL-LIEO personnel across both local network and live web environments.

System Testing Procedures

To satisfy Specific Objective 2 (testing and improving system performance, real-time cloud synchronization, and overall functionality), the researchers will systematically test the following specific features during the Alpha and Beta testing phases:

1. Ticket Generation & Sequence Integrity:

   - Atomic generation of unique, collision-free daily ticket sequence numbers (<PREFIX>-<MMDDYY>-<SEQ>) across concurrent requests.

   - Accurate classification of transaction types (New Business, Renewal, Retirement) and priority groups (Regular, Senior Citizen, PWD, Pregnant, Returning Client).

   - Dynamic QR code generation containing secure mobile tracking URLs and physical receipt formatting.

2. Predictive Waiting Time Mechanism (Phantom Queue Simulation):

   - Execution of the Phantom Simulation model injecting virtual tickets to determine true queue rank (R_true).

   - Application of dynamic abandonment discounting (D) based on queue length.

- Wait-time calculation accuracy evaluated against actual client service durations using Mean Absolute Error (MAE).

3. Adaptive Counter Allocation & Dynamic Priority Engine:

   - Real-time computation of the Smart Score S(t) incorporating base weights, wait-time aging, SLA panic bonus, skip bonus, and the zipper starvation-prevention rule.

   - Dynamic counter auto-balancing mode switching: transition between Peace Mode (all windows cater to all transactions) and Crunch Mode (proportional reallocation of windows based on queue congestion).

   - Starvation floor enforcement guaranteeing minimum staff coverage per active service.

4. Real-Time Cloud Synchronization & Offline Local Resilience:

   - WebSocket event latency: measuring end-to-end event delay between staff clicking "Call Next" and the public TV display updating (target latency: < 500 ms).

   - Cloud data synchronization: bidirectional syncing between the local SQLite database and Firebase Firestore.

   - Network fault tolerance: ensuring uninterrupted ticket issuance and counter calling on the local LAN during internet connection loss, followed by automatic delta re-synchronization once reconnected.

5. Automated Multi-Channel Notification Dispatch:

   - Automated SMS alert triggering when a client is within two spots of being called (NEAR_TURN).

- Automated SMS alert triggering upon initial counter call (CALLED) and counter re-call (RECALLED).

- Delivery reliability and latency of Web Push notifications on mobile client browsers.

6. Staff Queue Operations & Public Monitor Synchronization:

   - Frontline staff controls: Call Next, Recall, Skip/Defer, Postpone ("Return Tomorrow"), Mark No-Show, and Complete.

   - Audio-visual broadcast synchronization: real-time monitor display refresh, text-to-speech (TTS) voice announcements, and chime audio triggering.

Deploy and Review. In Agile, this final phase delivers a working software increment into the operational environment (LTS Group, 2023). The researchers will deploy the system in a hybrid cloud-hosted and local environment accessible by the Rosario BPL-LIEO. End-users (BPL-LIEO staff and visiting clients) will interact with the system during live trial runs and evaluate its quality using an evaluation questionnaire based on the ISO/IEC 25010 Software Quality Model, adapted from the study of Malaya et al. (2022).

The survey questionnaire (see Appendix C), adapted from the standardized instrument of Malaya et al. (2022), evaluates the system across all eight (8) product quality characteristics defined by the ISO/IEC 25010 Software Quality Model:

1. **Functional Suitability:** Functional completeness, functional correctness, and functional appropriateness in managing ticket generation, priority routing, and staff calling operations.
2. **Performance Efficiency:** Time-behavior (system response times under 500 ms), resource utilization, and throughput capacity during peak queue volume.
3. **Compatibility:** Co-existence and interoperability across heterogeneous client devices, local network routers, and cloud endpoints.
4. **Usability:** Appropriateness recognizability, learnability, operability, user error protection, user interface aesthetics, and accessibility across staff dashboards, TV displays, and mobile tracking screens.
5. **Reliability:** Maturity, service availability, fault tolerance, and data recoverability during local network or internet disconnections.
6. **Security:** Confidentiality of citizen contact information, integrity of queue records, non-repudiation, accountability, and user authentication via JWT and bcrypt.
7. **Maintainability:** Modularity, reusability, analyzability, modifiability, and testability of the backend controllers and database schemas.
8. **Portability:** Adaptability, installability, and replaceability across diverse web browsers (Chrome, Edge, Safari) and standard operating systems.

To ensure valid evaluations, characteristics are assigned based on user domain:

- End-Users (BPL-LIEO Staff and Visiting Citizens): Will evaluate the external, user-facing characteristics: Functional Suitability, Usability, Reliability, and Performance Efficiency.
- IT Experts / System Evaluators: Will evaluate all eight (8) characteristics, with specialized emphasis on the technical architecture: Security, Maintainability, Compatibility, and Portability.

The researchers will distribute the survey questionnaire to three (3) distinct user groups: all BPL-LIEO staff members who operate the ticket generator and staff dashboard, visiting citizens/clients who receive printed tickets and observe the public TV display, and IT Experts to assess the technical part of the system specifically in the security, maintainability, and portability. Because the visiting client population fluctuates daily, convenience sampling (n = 50) will be utilized for citizens, while total enumeration (N = 10) will be employed for the BPL-LIEO staff, and purposive sampling (n = 3) will be utilized for the IT Experts/System Evaluators.

Inclusion and Exclusion Criteria of Study Participants

To ensure that evaluation feedback accurately reflects operational realities and technical rigor, specific inclusion and exclusion criteria were established for each respondent group:

1. BPL-LIEO Personnel / Frontline Staff (n = 10)

- Inclusion Criteria:

- Regularly appointed or designated personnel, window counter operators, and administrative officers of the BPL-LIEO of LGU Rosario, La Union.
- Staff members actively assigned to process New Business, Permit Renewal, or Business Retirement transactions.
- Personnel who participated in system orientation and actively operated the Staff Dashboard, ticket calling, or analytics modules during trial runs.

- Exclusion Criteria:
  - Personnel assigned to other non-permitting municipal departments (e.g., Treasury, Assessor's, Mayor's Office, Engineering) who do not operate BPL-LIEO queuing windows.
  - Personnel on prolonged medical or administrative leave during the evaluation period.
- Rationale: Only personnel directly handling permit workflows can objectively evaluate the usability, counter-allocation efficiency, and operational relevance of the staff interface.

2. Transacting Citizens and Business Representatives (n = 50)

- Inclusion Criteria:
  - Individuals aged 18 years or older physically transacting at the BPL-LIEO for business registration, renewal, or closure.

- Clients issued an official queue ticket across Regular, Senior Citizen, PWD, Pregnant Women, or Returning Client classifications.
- Clients who observed the public TV display and/or utilized a smartphone to scan the ticket QR code for mobile queue tracking and SMS alerts.

- Exclusion Criteria:
  - Minors (individuals below 18 years of age) to uphold ethical and informed consent standards.
  - Companions, drivers, or relatives who merely accompanied the transacting client and did not personally undergo the queuing process.
  - Citizens visiting the municipal hall for unrelated government transactions or generic inquiries outside the BPL-LIEO.
- Rationale: Non-transacting companions and visitors of other offices do not experience the queuing system, so excluding them eliminates invalid or speculative survey responses.

3. IT Experts / System Evaluators (n = 3)

- Inclusion Criteria:
  - Professionals holding at least a Bachelor's or Master's degree in Computer Science, Information Technology, or closely related computing discipline, or with at least three (3) years of industry

experience in software engineering, web application development, or system administration.

- Demonstrable background in evaluating web architectures, relational/cloud databases, and network communication protocols.

- Exclusion Criteria:
  - Individuals lacking formal computing education or industry software experience.
  - Proponents and individuals directly involved in the system's development, avoiding conflicts of interest.
- Rationale: Evaluating non-functional ISO/IEC 25010 characteristics—particularly Security, Maintainability, Reliability, and Portability—requires specialized technical competencies beyond the knowledge of ordinary end-users.

The citizen population ($N = 150$) represents the projected aggregate foot-traffic of transacting business representatives visiting the BPL-LIEO over the designated 5-day live beta-testing deployment period (an average of 30 clients per day), from which a representative sample of $n = 50$ clients will be drawn via convenience sampling.

Table 1. Distribution of Respondents

Respondents	Total Population N	Sample Population n
BPL-LIEO Staff	10	10
Citizens / Clients	150	50
IT Experts/ System Evaluators	3	3
TOTAL	163	63

Data Gathered

The researchers will gather information on the current state of service delivery and workflow in the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO). Assessment of system's quality will also be conducted through a survey using the ISO-Based questionnaire of Malaya et al. (2022) to both the BPL-LIEO staff and clients.

Data Analysis

To analyze the data gathered from the survey questionnaires, the formula for the weighted mean will be used to evaluate the web-based queuing management system with predictive waiting mechanism and adaptive counter allocation. This statistical tool is

appropriate for determining the average response of the participants based on a 5-point Likert Scale. Below presents the average score range and the corresponding descriptive equivalent rating. To address Specific Objective 3, data gathered from the ISO/IEC 25010 evaluation questionnaires will be analyzed using the weighted mean. The grand mean will be computed for each of the eight (8) quality characteristics as well as the overall composite mean to determine the software quality level according to the established 5-point Likert scale criteria.

Scale	Mean Score	Descriptive Equivalent Rating
1 (Strongly Disagree)	1.00 – 1.79	Poor
2 (Disagree)	1.80 – 2.59	Fair
3 (Neutral)	2.60 – 3.39	Good
4 (Agree)	3.40 – 4.19	Very Good
5 (Strongly Agree)	4.20 – 5.00	Excellent

Responses with a score ranging from 3.40 to 5.00 indicate varying levels of agreement. Specifically, scores between 4.20 and 5.00 (Strongly Agree) indicated that the system's quality is excellent. Scores between 3.40 and 4.19 (Agree) suggested that the system quality is very good, while scores between 2.60 and 3.39 (Neutral) indicated that the system's quality is good. Conversely, responses falling below 2.60, signifies client dissatisfaction.

Aside from the Likert scale evaluation for user satisfaction, the technical performance of the system will be analyzed using the following metrics:

- **Summary of Alpha and Beta Testing Results:** The results of the core functionality tests (e.g., QR scanning, ticket generation, SMS delivery) will be analyzed using a Pass/Fail Matrix. The percentage of successful test cases against expected outputs will be calculated. Any failed test cases will be isolated, logged, and analyzed for root-cause resolution in the subsequent sprint.
- **Real-Time Cloud & Local Synchronization Performance:** Network performance will be analyzed by measuring the Latency (in milliseconds) of WebSocket events—specifically the time delay between a staff member clicking "Call Next" and the public TV display updating. Furthermore, Data Consistency will be analyzed by running a cross-reference query between the local SQLite database and the Firebase Cloud database to ensure 0% data loss during internet interruptions.
- **Predictive Wait Time Accuracy Analysis:** The precision of the Phantom Queue Simulation algorithm will be statistically analyzed using Mean Absolute Error (MAE). During live testing, the system will record the *Predicted Wait Time* of every ticket and compare it against the *Actual Wait Time* upon transaction completion. The MAE will calculate the average deviation in minutes, providing a mathematically sound validation of the system's accuracy.

Ethical Considerations

In adherence to the Data Privacy Act of 2012 (Republic Act No. 10173) and institutional ethical research guidelines of Don Mariano Marcos Memorial State University,

the researchers formulated a comprehensive Data Management Plan covering the ethical handling, storage, retention, and disposal of all data collected during system development, testing, and evaluation:

1. Informed Consent and Voluntary Participation

   All survey participants (BPL-LIEO frontline staff, transacting citizens, and IT experts) will be informed of the nature, objectives, and academic scope of the study. Participation in the ISO/IEC 25010 evaluation is entirely voluntary, and respondents may withdraw at any time without penalty or denial of public service. Written or digital informed consent will be obtained prior to questionnaire administration, and all survey feedback will remain strictly anonymous and confidential.

2. Data Management Plan

   The system processes two categories of information:

   a. Citizen Transaction Data: Includes generated queue ticket numbers, selected transaction types (New, Renewal, Retirement), demographic priority tags (Regular, Senior Citizen, PWD, Pregnant, Returning), and mobile phone numbers voluntarily provided for SMS notifications. No sensitive financial documents, tax declarations, or payment credentials are collected or processed by the queuing system.

   b. Operational and Audit Data: Includes staff user accounts, timestamps of counter operations (call, recall, completion), queue duration logs, and counter allocation history used exclusively for operational orchestration and statistical reporting.

3. Data Storage and Security

   All transactional queue records and user accounts will be securely persisted on-premise using a local relational database (SQLite via Prisma ORM) hosted within the municipal hall's local server environment. For external mobile tracking, synchronized queue states are transmitted securely over TLS/HTTPS encryption to cloud-hosted database collections (Firebase Firestore) governed by strict role-based access rules. User credentials and staff passwords are cryptographically hashed using bcrypt, and API interactions are authorized through JSON Web Tokens (JWT). Direct database access is restricted exclusively to authorized system administrators and designated BPL-LIEO personnel.

4. Data Retention Policy

   To minimize privacy exposure:

   a. Transient Citizen Contact Details: Mobile phone numbers collected exclusively for SMS dispatch are retained in active memory and operational storage only for the duration of the operating business day. Once a transaction is marked as completed or closed at the end of the day, client contact numbers are automatically masked or decoupled from queue records.

   b. Historical Analytics Logs: De-identified transaction records (containing only sequence numbers, transaction categories, priority classifications, wait times, and service durations) are retained for a period of one (1) fiscal year to facilitate comparative Year-over-Year (YoY) administrative reporting and seasonal volume forecasting for the BPL-LIEO.

5. Data Disposal

   Upon the expiration of the defined retention period or upon the final academic conclusion of the thesis project, all temporary testing datasets and local backup snapshots containing identifiable user logs will be permanently deleted using secure cryptographic file shredding and database truncation procedures. Physical survey evaluation sheets collected from participants will be shredded and incinerated in compliance with university research data disposal protocols.

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APPENDICES

APPENDIX A

Letter to the Adviser

Don Mariano Marcos State University
South La Union Campus
COLLEGE OF COMPUTER SCIENCE
Agoo, La Union

July 1, 2026

EMELY A. MUNAR, DIT (c)
Faculty Member, College of Computer Science
DMMMSU – SLUC
Agoo, La Union

Sir/Madam:

We, the undersigned BS Computer Science students are enrolled in the subject Thesis Writing. One of the requirements of the course is for us to undergo the process of research and be able to create a project that will beneficial to the university.

In connection to this, we would like to ask for your support to be our **Thesis Adviser** for our study entitled *"Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union"*.

We hope that this request will merit your approval. We will highly appreciate your considerations to this regard.

Thank you very much and more power.

Very truly yours,

TANDAS, ROJEAN F.
Leader

BARBOSA, MANRIJAN M.
Member

ESTRELLES, CARREN NICOLE A.
Member

GARCIA, JOHN DAVID R.
Member

RULLODA, JADE ALLYSA V.
Member

Approved:

EMELY A. MUNAR, DIT (c)
Thesis Adviser

APPENDIX B

Interview Guide Questions

Target Interviewee: BPL-LIEO Head / Authorized Personnel

Objective: To gather operational requirements, identify current queuing bottlenecks, and determine the necessary features for the proposed automated queuing system.

Part I: Current Workflow and Manual Processes

1. Can you describe the step-by-step process of how clients are currently accommodated from the moment they enter the BPL-LIEO office?
2. How do you currently distinguish and manage clients applying for a *New Business Permit* versus those processing a *Renewal*?
3. Without an automated system, how are queue numbers or the order of service currently assigned, tracked, and announced to waiting clients?

Part II: Pain Points and Operational Bottlenecks

4. What are the most common difficulties or bottlenecks your staff experience with the current manual queuing method during peak seasons?
5. How do you currently handle situations where a client is called for their turn but is not present at the waiting area?
6. How are "Return Tomorrow" or incomplete transactions currently managed, and how do you prioritize these clients when they return the following day?

Part III: Proposed System Features and Requirements

7. If an automated proposed web-based queuing management system is implemented, what specific information would be most helpful to display on the public TV monitor for the waiting citizens?

8. What specific controls or functions would your frontline staff need on their computer dashboard to effectively manage the flow of clients (e.g., calling the next number, skipping absent clients, marking transactions as done)?
9. What kind of data or statistical reports would be most useful for the BPL-LIEO administration to monitor daily transactions and evaluate office performance?

APPENDIX C

Evaluation Questionnaire based on ISO/IEC 25010

(Malaya, et al. 2022)

Name (Optional):_____ Date:_____

User Type: [ ] BPL-LIEO Staff [ ] Citizen / Client

Instructions: Please evaluate the Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation using the following scale:

5 - Strongly Agree
4 - Agree
3 - Neutral
2 - Disagree
1 - Strongly Disagree

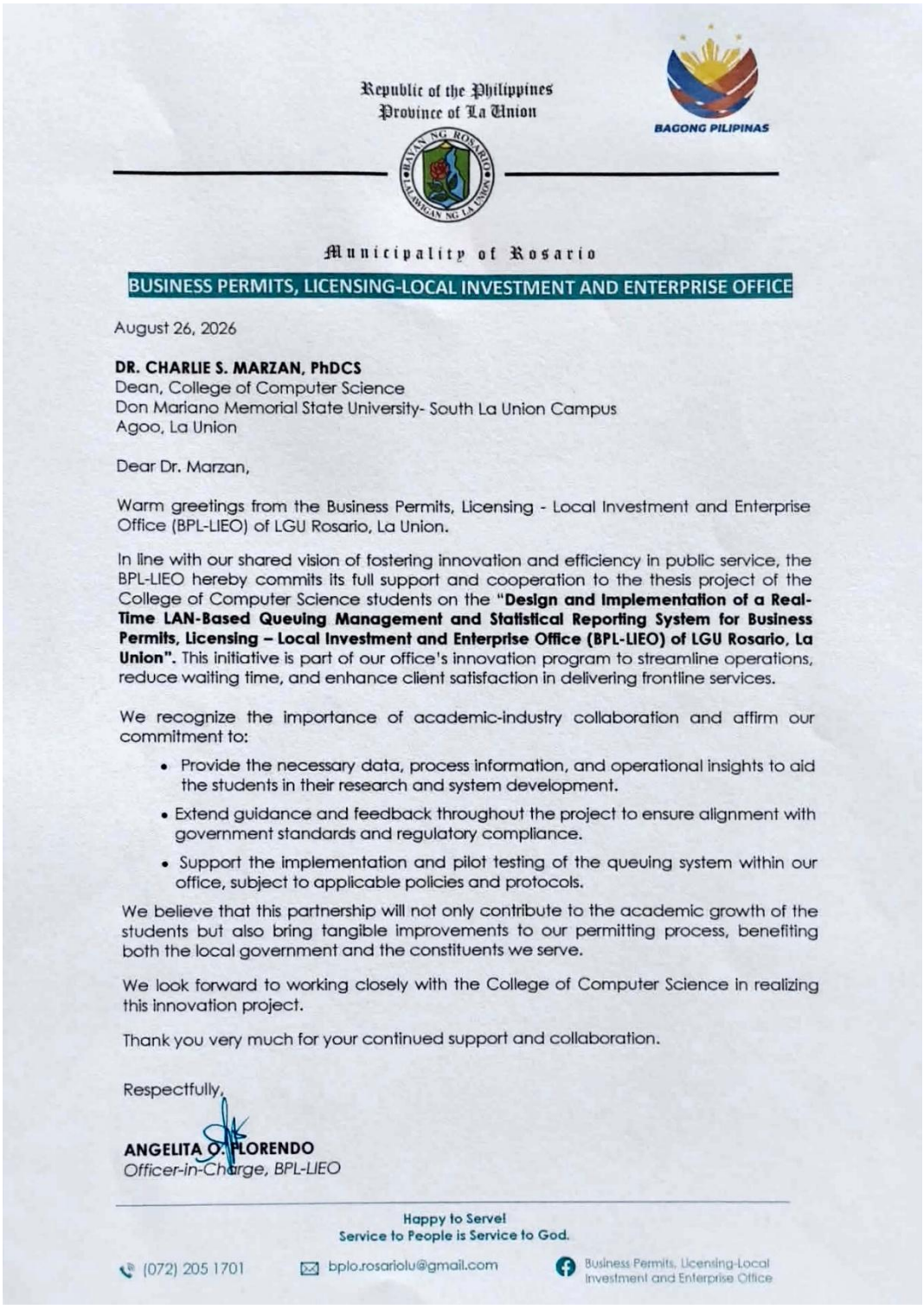
FUNCTIONALITY	**5**	**4**	**3**	**2**	**1**
1. Services covered all the specific tasks and user objectives, which were designed. (functional completeness)					
2. Services provided the correct results with the needed degree of precision. (functional correctness)					
3. Services are designed to facilitate accomplishment of specified tasks, more precisely, the execution of a business process. (functional appropriateness)					
RELIABILITY	**5**	**4**	**3**	**2**	**1**
1. Whenever a service consumer requests some information, it is expected that a response is returned. (maturity)					
2. Services are available when they are requested. (availability)					
3. Services can create strategies that may be performed when a failure happens on some hardware or software. (fault tolerance)					

4. Services have the ability to recover data when some interruptions or failures occur. (recoverability)					
USABILITY	**5**	**4**	**3**	**2**	**1**
1. Users can recognize whether this service is appropriate for their needs through service description that relate information such as service functionality and data types transmitted. (appropriateness recognizability)					
2. Service can facilitate the understanding of its operation. (learnability)					
3. Service has a document that allows exchange messages between services. (operability)					
4. The structure of a service does not allow making errors from wrong inputs. (user error protection)					
5. User interface aesthetics					
6. Accessibility					
EFFICIENCY	**5**	**4**	**3**	**2**	**1**
1. Time spent by a service to process a request and return a response. (time behavior)					
2. Services use resources such as servers to access information of other applications. (resource utilization)					
3. Service has the ability to remain working even with large number of accesses at the same time. (capacity)					
MAINTAINABILITY	**5**	**4**	**3**	**2**	**1**
1. It hosts a network accessible software module. (modularity)					
2. Services are reusable. (reusability)					
3. Analyze change impact when services need to be modified. (analyzability)					
4. Services are loosely coupled. This characteristic reduces the dependency between services, increasing modifiability. (modifiability)					

5. Services can be tested, for instance, through automated tools for functional testing. (testability)					
PORTABILITY	**5**	**4**	**3**	**2**	**1**
1. Although web/LAN services run remotely on a server, it can happen in a change of platform. (adaptability)					
2. Although web/LAN services run remotely on a server, it can happen in a change of platform. (installability)					
3. Although web/LAN services run remotely on a server, it can happen in a change of platform. (replaceability)					
COMPATIBILITY	**5**	**4**	**3**	**2**	**1**
1. Different composite services can share the use of same service operations. (co-existence)					
2. Services allow interaction between systems through the use of interfaces and communication protocols. (interoperability)					
SECURITY	**5**	**4**	**3**	**2**	**1**
1. Information shared can be accessed only to an authorized service client. (confidentiality)					
2. Services are developed to prevent unauthorized access to, or modification of private data. (integrity)					
3. Service provider constructs strategies to prove that an information has been delivered to a service consumer. (non-repudiation)					
4. Services are autonomous. (accountability)					
5. The identity of the external service provider is authenticated. (authenticity)					

APPENDIX D

Commitment Letter



Republic of the Philippines
Province of La Union

BAGONG PILIPINAS

Municipality of Rosario

BUSINESS PERMITS, LICENSING-LOCAL INVESTMENT AND ENTERPRISE OFFICE

August 26, 2026

DR. CHARLIE S. MARZAN, PhDCS
Dean, College of Computer Science
Don Mariano Memorial State University- South La Union Campus
Agoo, La Union

Dear Dr. Marzan,

Warm greetings from the Business Permits, Licensing - Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union.

In line with our shared vision of fostering innovation and efficiency in public service, the BPL-LIEO hereby commits its full support and cooperation to the thesis project of the College of Computer Science students on the **"Design and Implementation of a Real-Time LAN-Based Queuing Management and Statistical Reporting System for Business Permits, Licensing – Local Investment and Enterprise Office (BPL-LIEO) of LGU Rosario, La Union".** This initiative is part of our office's innovation program to streamline operations, reduce waiting time, and enhance client satisfaction in delivering frontline services.

We recognize the importance of academic-industry collaboration and affirm our commitment to:

- Provide the necessary data, process information, and operational insights to aid the students in their research and system development.
- Extend guidance and feedback throughout the project to ensure alignment with government standards and regulatory compliance.
- Support the implementation and pilot testing of the queuing system within our office, subject to applicable policies and protocols.

We believe that this partnership will not only contribute to the academic growth of the students but also bring tangible improvements to our permitting process, benefiting both the local government and the constituents we serve.

We look forward to working closely with the College of Computer Science in realizing this innovation project.

Thank you very much for your continued support and collaboration.

Respectfully,

ANGELITA O. FLORENDO
Officer-in-Charge, BPL-LIEO

Happy to Serve!
Service to People is Service to God.

(072) 205 1701 bplo.rosariolu@gmail.com Business Permits, Licensing-Local Investment and Enterprise Office

CURRICULUM VITAE



ROJEAN F. TANDAS
Casilagan, Sto. Tomas, La Union
+63 994 742 2028
rtandas23109793@student.dmmmsu.edu.ph

:

Personal Information

Birthdate: July 22, 2005
Place of Birth: Rosario, La Union
Gender: Male
Age: 21
Nationality: Filipino
Religion: Roman Catholic
Civil status: Single
Father's Name: Bencio M. Tandas
Mother's Name: Maribel F. Tandas

Educational Background

2023-present	**College Level** Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union *Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation*
2021-2023	**Senior High School Level** Santo Tomas National High School Patac, Santo Tomas, La Union *With Honors*

2016-2021	**Junior High School Level** Santo Tomas National High School Patac, Santo Tomas, La Union
2011-2016	**Elementary Level** Casilagan Elementary School Casilagan, Santo Tomas, La Union

Special Skills and Talents

Technical Skills

- Programming Languages: C, JavaScript
- Web Technologies: HTML / HTML5
- System & Environment Management: Software Installation, OS Configuration, Basic Troubleshooting
- Productivity Software: Microsoft 365 (Word, Excel, PowerPoint)

Capabilities, Strengths & Core Aptitudes

- Technical Troubleshooting: Ability to configure system environments, install essential developer tools, and resolve baseline software issues.
- Foundational Algorithmic Thinking: Understanding of procedural programming concepts, control structures, and syntax logic via C.
- Front-End Development Basics: Structuring clean, semantic web layouts and implementing foundational client-side scripting.
- Documentation & Reporting: Creating structured technical write-ups, presentations, and spreadsheet records.

I hereby certify that the above statements are correct to the best for my knowledge.

RTandas

ROJEAN F. TANDAS

Signature over printed name



CARREN NICOLE A. ESTRELLES
Namonitan, Santo Tomas, La Union
+63 967 342 6275
cnestrelles22111243@student.dmmmsu.edu.ph

:

Personal Information

Birthdate: August 21, 2004
Place of Birth: Santo Tomas, La Union
Gender: Female
Age: 22
Nationality: Filipino
Religion: Born Again Christian
Civil status: Single
Father's Name: Edward C. Estrelles
Mother's Name: Marites A. Estrelles

Educational Background

2022-present	**College Level** Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union *Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation*
2020-2022	**Senior High School Level** Polytechnic College of La Union San Nicolas Sur, Agoo, La Union
2016-2020	**Junior High School Level** Polytechnic College of La Union San Nicolas Sur, Agoo, La Union With Honors
2010-2016	**Elementary Level**

Fort Del Pilar Elementary School
Philippine Military Academy, Baguio City

Special Skills and Talents

Technical Skills

- Web Development: HTML5, CSS3, JavaScript (ES6)
- Hardware & Systems: Computer Hardware Diagnostics, Component Assembly, System Maintenance & Upgrades

Capabilities, Strengths & Core Aptitudes

- Front-End Fundamentals: Developing structured, styled, and interactive user interfaces using modern client-side web technologies.
- Hardware Troubleshooting: Identifying component-level faults, handling peripheral integration, and performing routine hardware maintenance.
- Systems Familiarity: Understanding the interplay between physical hardware architecture and software execution.

I hereby certify that the above statements are correct to the best for my knowledge.



CARREN NICOLE A. ESTRELLES
Signature over printed name



JADE ALLYSA V. RULLODA
Poblacion, Aringay, La Union
+63 991 306 3648
jarulloda23200033@student.dmmmsu.edu.ph

:

Personal Information

Birthdate: September 20, 2004
Place of Birth: San Fernando, La Union
Gender: Female
Age: 21
Nationality: Filipino
Religion: Roman Catholic
Civil status: Single
Father's Name: Wilson D. Rulloda
Mother's Name: Susan V. Rulloda

Educational Background

2023-present	**College Level** Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union *Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation*
2021-2023	**Senior High School Level** Aringay National High School Poblacion, Aringay, La Union With Honors
2017-2022	**Junior High School Level** Aringay National High School Poblacion, Aringay, La Union With Honors
2011-2016	**Elementary Level**

Aringay Central Elementary School
Poblacion, Aringay, La Union

Special Skills and Talents

Technical Skills

- Programming Languages: Java
- Web Technologies: HTML / HTML5
- Core Computing: Environment Setup, File Management, System Operations

Capabilities, Strengths & Core Aptitudes

- Object-Oriented Programming (OOP) Fundamentals: Understanding foundational OOP principles, class structures, and programmatic logic in Java.
- Web Content Structuring: Building semantic, accessible document outlines and baseline web structures using HTML.
- Technical Adaptability: Quickly learning syntax structures, core computing workflows, and essential developer environments.

I hereby certify that the above statements are correct to the best for my knowledge.

JADE ALLYSA V. RULLODA
Signature over printed name



JOHN DAVID R. GARCIA
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+63 962 526 9054
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Personal Information

Birthdate: November 3, 2004
Place of Birth: Dagupan City, Pangasinan
Gender: Male
Age: 21
Nationality: Filipino
Religion: Born Again Christian
Civil status: Single
Father's Name: Victoriano S. Garcia
Mother's Name: Marites R. Garcia

Educational Background

2023-present	**College Level** Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union *Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation*
2021-2023	**Senior High School Level** Graystone Institute of the Philippines 3F Maradel Bldg., AB Fernandez Street, Dagupan City, Pangasinan With High Honors
2016-2021	**Junior High School Level** San Fabian National High School Cacapit Street, San Fabian, Pangasinan With Honors
2011-2016	**Elementary Level** Tocok Elementary School

Tocok, San Fabian, Pangasinan
With Honors

Special Skills and Talents

Technical Skills

- Languages & Web Stack: Python, Java, JavaScript, React, Node.js, Express, Tailwind CSS, Semantic HTML/CSS.
- Systems & Environment: Linux (Ubuntu) Administration, Git/GitHub, Supabase, CLI Operations.
- Architecture & Design: UML Modeling, MVP Feature Scoping, Figma (UI Mockups & Structural Site Maps).

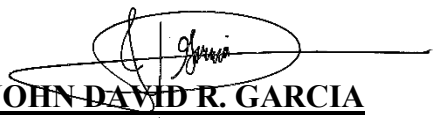
Capabilities, Strengths & Core Aptitudes

- AI Prompt Engineering: Crafting highly specific master prompts, custom instructions, and behavioral parameters to optimize coding and technical workflows.
- System & OS Optimization: Diagnosing and safely debloating mobile operating systems via precise system package management.
- Product & Business Strategy: Formulating Business Model Canvases, transaction flows, and operational workflows for conceptual software platforms.

Leadership & Soft Skills

- Public & Academic Governance: Managing public funds and community initiatives as an SK Treasurer, alongside logistical coordination and peer facilitation as a Class Secretary and Captain.
- Technical Communication: Translating system architectures into formal client proposals and delivering structured presentations on programming philosophies.

I hereby certify that the above statements are correct to the best for my knowledge.



JOHN DAVID R. GARCIA
Signature over printed name



MANRIJAN M. BARBOSA
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Personal Information

Birthdate: June 19, 2004
Place of Birth: Baguio City
Gender: Male
Age: 22
Nationality: Filipino
Religion: Roman Catholic
Civil status: Single
Father's Name: Manrigold E. Barbosa
Mother's Name: Florence M. Barbosa

Educational Background

2023-present	**College Level** Bachelor of Science in Computer Science Don Mariano Marcos Memorial State University Consolacion, Agoo, La Union *Web-Based Queuing Management System with Predictive Waiting Mechanism and Adaptive Counter Allocation*
2020-2022	**Senior High School Level** Rosario Integrated School Poblacion East Rosario La Union
2016-2020	**Junior High School Level** Saint Mary's Academy Poblacion East Rosario La Union
2010-2016	**Elementary Level**

Rosario Integrated School
Poblacion East Rosario La Union

Special Skills and Talents

Technical Skills

- Software Development: Core Programming Fundamentals, Code Logic, and Syntax Application
- Data & Analytics: Data Analysis, Information Processing, and Analytical Modeling
- Design & Multimedia: Graphic Design, Digital Asset Creation, and Visual Communication

Capabilities, Strengths & Core Aptitudes

- Analytical Problem-Solving: Applying structured methodologies to identify, assess, and resolve complex technical and logistical challenges.
- Technical Troubleshooting: Diagnosing system errors, debugging code logic, and implementing effective, practical solutions.
- Dynamic Adaptability: Quickly acclimating to new technologies, development environments, and shifting project requirements.
- Effective Communication: Articulating technical concepts clearly to both technical peers and cross-functional team members.

I hereby certify that the above statements are correct to the best for my knowledge.

MANRIJAN M. BARBOSA
Signature over printed name